

Solution Requirements

Date	27 June 2026
Team ID	
Project Name	Rising Waters: AI-Powered Flood Risk Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Weather Data Input	The system allows users to enter rainfall data (Annual, Monsoon, Pre-Monsoon, Post-Monsoon) and cloud visibility for a specific region.	High
2	Input Validation	The system validates all entered values to ensure they are complete, numeric, within acceptable ranges, and not negative before processing the prediction.	High

Step 1: Functional Requirements (FR) (Continued)

S.No	Requirement Category	Requirement Description	Priority
3	Flood Risk Prediction	The system uses a trained machine learning model to predict the flood risk based on the provided inputs.	High
4	Risk Classification	The system classifies the prediction into Low, Moderate, or High flood risk categories.	High
5	Probability Display	The system displays the probability (in %) of flood occurrence for the given inputs.	High
6	Advisory Generation	The system provides appropriate advisory messages based on the predicted risk level.	High
7	User Interface	The system provides a simple, responsive, and user-friendly web interface for data entry and results display.	Medium
8	Results Summary	The system displays and summarizes the prediction results in an easy-to-understand format.	Medium
9	Error Handling	The system shows clear error messages for invalid inputs and guides users to correct them.	Medium
10	Model Management	The system loads the latest trained model and metadata automatically at startup.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should process user input and display the flood risk prediction without noticeable delay.	Prediction displayed within 2 seconds
2	Scalability	The system should support additional regions, large datasets, and increased usage without major changes to the application.	Supports future feature and dataset expansion
3	Security & Data Privacy	User inputs should be validated, and the system should process data securely without storing personal information unnecessarily.	100% input validation with secure data handling
4	Reliability & Availability	The application should consistently generate accurate risk predictions and remain stable during normal operation.	99% successful prediction rate during testing
5	Usability & Accessibility	The interface should be simple, intuitive, and easy to use for disaster management teams with basic computer knowledge.	Users can complete prediction in 3 simple steps
6	Maintainability & Portability	The project should have a modular code structure and run on Windows, Linux, and macOS with minimal configuration.	Cross-platform support with modular Python code